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A High-Resolution Reddening Map Toward the Southern Galactic Bulge from Red Clump Giants

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Abstract. Using the colors of red clump giants, we construct a reddening map with 2-arcminute resolution covering 40 square degrees toward the southern bulge. Toward the Kestrel-7 field we find E(B−V) = 0.08 ± 0.02, and we recommend $A_I/E(B-V) = 1.86$ for standard $R_V = 3.1$ extinction.

1. Method

Red clump giants have a narrow intrinsic color distribution, so their observed color excess traces line-of-sight reddening. We bin stars spatially and fit the clump color in each bin.

2. Results

Reddening varies by up to 0.05 mag on arcminute scales in this region. The Kestrel-7 field is comparatively uniform.

Table 1. Summary of key values.

Field	E(B−V)	σ
Kestrel-7	0.08	0.02
Field B-12	0.21	0.03
Field C-04	0.15	0.02

References

Haddad, R., Tanaka, S., & Ruiz, M. 2021, Proc. Mock Astron. Soc., 48, 233